HONGKUN HE

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EDUCATION

University of Southern California Aug 2026 - Present
M.S. in Computer Science

The Pennsylvania State University Sep 2022 - May 2026
B.S. in Computer Science (Engineering) | Minor in Cybersecurity | PA, USA

Relevant Coursework: Computer Organization and Design, Discrete Mathematics, Digital Design, Data Structures and Algorithms, Probability, Communication Networks, Programming Language Concepts

TECHNICAL SKILLS

Languages	Python, C, Scheme
AI / LLMs	Transformers, LoRA, PEFT, SFT, Hugging Face, LLaMA-Factory
LLM Systems	RAG, embeddings, vector databases, retrieval optimization, prompt engineering
Agents	ReAct, tool calling, multi-step reasoning
Data	Pandas, feature engineering, data cleaning, data visualization

EXPERIENCE & PROJECTS

Shanghai MoonSeer Technology Co., Ltd. **Algorithm Engineer** May-Aug 2026
Shanghai, China

- Developed an indoor navigation agent for visually impaired users using multi-frame visual understanding, semantic map matching, and LLM-based decision-making, with local-planning, voice, and haptic feedback integration.
- Designed structured prompts, schema validation, and exception-handling protocols to map model outputs into Arrow directions and PlanningGuidance.
- Built multi-frame visual feature matching and coordinate-free topological trajectories for image and task-state management.
- Built log-replay and prompt-debugging tools and automated regression tests for agent modules.
- Architected TaskManager and NavigationSubAgent; split navigation into Map Loop, Nav Task Loop, and Fast Loop for environment understanding, task-stage control, and long-term context management.
- Optimized KV-cache usage and the model-call pipeline, improving response speed by 32.6%.

Shanghai Hojo Intelligence Technology Co., Ltd. **Algorithm Intern** May-Aug 2025
Shanghai, China

- Designed and implemented an LLM-powered in-vehicle voice interaction system spanning navigation, vehicle control, and entertainment scenarios.
- Built a prompt and intent-parsing framework that improved intent-recognition accuracy by 22% across 10+ in-vehicle scenarios and reduced command failures.
- Converted 10K+ raw conversation logs into structured multi-turn training data, improving data usability by 35%.
- Fine-tuned models with LoRA (PEFT), reducing training cost by 40% under limited compute resources.
- Developed an ADB-based GUI automation pipeline that generated 1K+ state-action-semantics training samples.
- Reduced inference latency by 18% through prompt optimization and caching.

Taproot Kitchen Inventory & Cost Management System **Algorithm Team Lead** Jan-May 2026
Pennsylvania, USA

- Led development of a production-deployed cloud inventory and recipe-cost management system for a nonprofit organization.
- Extended the InvenTree Django backend to support recipe-cost modeling and automated BOM cost calculation.
- Optimized relational models and indexes, improving ingredient-recipe query performance by 40%.
- Deployed on AWS EC2 with Docker, including persistent storage, environment isolation, and reverse-proxy configuration.
- Implemented role-based access control (RBAC) for multi-user permission isolation.
- Automated cost calculations, reducing manual reconciliation workload by 50% and enabling real-time cost visualization.